

Introducing Shopper Avatars in a Virtual Reality Store

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Figure 1: Virtual Store environment – the purchasable potato chips product (left). A participant in the VR store – the user remained in a stationary position while using the controller to navigate the VR store via Teleportation locomotion (right).

ABSTRACT

Immersive Virtual Reality (iVR) is rapidly altering the world we live in. Studies have shown that people exhibit similar behaviours and reactions in virtual retail scenarios, thus making iVR a promising tool for researchers to study in-store shopper behaviour. This paper investigates whether virtual avatars can affect users' perceived telepresence and shopping behavior. We recruited 54 participants to experience the virtual store and our findings showed the mere presence of virtual avatars in the store had no significant effect on participants' sense of telepresence and shopping behaviour. We believe this research highlighted the need for further study and contributed to the development of fully-fledged virtual simulated shopping experiences.

Index Terms: Virtual store—Virtual Reality—Telepresence—Avatar—Shopping behaviour

1 INTRODUCTION

Immersive Virtual Reality (iVR) is rapidly altering the world we live in. Studies demonstrated that visualisations experienced in immersive VR can be used effectively in various domains [1, 11]. iVR has seen increasing adoption thanks to the advent of affordable consumer head-mounted devices and beckons with significant potential for influencing users' attitude and behaviour [7]. To obtain a full sense of telepresence in the environment, iVR technology is gradually committed to creating perceptions through more realistic images, sounds or other stimuli surrounding the user in the iVR environment [10]. Thereby iVR technology, mediated by factors such as telepresence and social presence, can create virtual worlds that resemble the physical world to a greater extent than traditional media could. As a result, one can expect greater similarities between user behaviour in the virtual environment and their behaviour in the real world [4]. Thus, behaviour in immersive virtual environments can serve as a close proxy for real human behaviour, which makes iVR a promising research tool for human behaviour research [3].

Social cues and interactions with other shoppers are a natural part of any shopping experience in a bricks-and-mortar store [5]. In a

virtual context, the perception of others in form of anthropomorphic avatars in a virtual environment has been shown to have a direct impact on a user's sense of presence [6]. This finding might have implications for the perceived pleasantness of iVR store environments and observed shopper behaviour. The relevance of the perception of other shoppers in a virtual store context thus provides the impetus for the current study.

In this paper, we investigated the effectiveness of introducing avatars on shoppers' perceived telepresence and the consequent impact on observed shopper behaviour. We investigated two research questions: 1) does the presence of virtual shopper avatars in the store would influence users' perceived telepresence?; 2) would this consequently lead to an impact on shopping behaviour, measured by changes in shopping trip duration?

2 RESEARCH DESIGN AND EVALUATION

We created an immersive virtual simulated store using Unreal Engine 4 based on an Australasian retail store. The presence/absence of virtual avatars was set as the independent variable in this research. We populated the virtual store with four anthropomorphic avatars (two females and two males) which had basic body animations including walking along a randomly generated path through the virtual store. To experience the virtual store, participants were equipped with a Virtual Reality headset and asked to enter and explore the virtual environment via a controller-supported instant teleportation movement technique. The iVR store included a total of four merchandise shelves which were placed in the center of the room. Except for one purchasable potato chips brand products that were displayed on the shelves, the rest products were simple static meshes that could not be picked up to minimise the processing load of the application.

The research was approved by Massey University's Human Ethics Committee. We recruited 54 participants (23 self-identified as females, 31 self-identified as males), staff and students of Massey University and The University of Auckland. To answer our research questions, we randomly divided the participants into two groups: Avatar (n=27) and non-avatar (n=27). The experimental procedure followed the procedure outlined by Schnack et al. [8]. Participants in both experimental groups received the same task instructions before starting the main experiment. The task was to find a specific brand of BBQ-flavoured potato chips in the virtual store (see Fig. 1(left)), put one or more in the shopping basket, walk to the counter and complete the purchase by placing the shopping basket on the check-out counter. While the avatar group found the virtual store populated

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Table 1: Independent Samples t-Test - Store observations

	Non-avatar Mean	SD	Avatar Mean	SD	Sig.*
Time spent (in secs)	192.31	88.93	199.66	94.19	0.769

*t-test (2-tailed)

with virtual shopper avatars, the non-avatar group did not find any other shoppers in the store. Fig. 1(right) shows a participant interacting with the VR environment. A questionnaire related to the participant's personal information and the VR experience was given to each participant after the experiment.

3 RESULTS

A total of 54 valid responses for the questionnaire were obtained and analysed using the IBM SPSS Package. To allow hypothesis testing in this experiment, we applied questionnaire items that were measured on a Five-point Likert-Scale ranging from 1='Strongly disagree' to 5='strongly agree'. The questions covered the investigated user experience constructs telepresence, and the instrument was adopted from previous research [9].

Looking at differences between the two experimental groups to investigate RQ1, a comparison of questionnaire item means only found minor differences between both conditions. The non-avatar group with a mean of 3.57 out of 5 showed slightly higher scores than the avatar group with a mean of 3.38. However, the t-test indicated that the differences were not statistically significant with a p-value of .638 for telepresence. The overall lack of statistically significant differences between the avatar and non-avatar group indicate that the use of virtual avatars did not impact participants' performance scores for the investigated construct.

We also looked at behavioural observations, namely the total time spent in the VR store in avatar vs non-avatar condition (in response to RQ2), as shown in Table 1. The results showed that participants in the experimental condition (with avatar), in fact, spent on average more time (199.66 secs), compared to the control condition (non-avatar: 192.32 secs). However, a t-test did not find any statistically significant differences between the two groups. Thus, there is no effect of avatar presence on the observed behavioural metric of shopping time.

4 CONCLUSION AND FUTURE WORK

The main objectives of this study were to investigate whether the presence/absence of avatars in the virtual store would influence shoppers' sense of telepresence and in-store behaviour. Our findings show that the mere presence of virtual avatars does not result in any significant effect on investigated presence measures - namely participants' sense of telepresence. Thus, the assumption that simply adding virtual avatars to a virtual store environment would increase presence to influence user's behaviour has to be studied further.

While our work provides first evidence about the impact of shopper avatars and VR store user perception and behaviour, the current study comes with some limitations that leave room for further research in this area. Firstly, except for one purchasable potato chips brand product that was displayed on the shelves, the rest products were simple static meshes that could not be picked up. In future studies, user's should be allowed to interact with all products to better experience the social context in the virtual store. We also recommend further research in this field with the aim of examining the effectiveness of smart avatars on participants' sense of presence and shopping behavior as previous studies demonstrated that improvements to an avatar's behavioural realism can significantly improve perceived co-presence [2]. Artificial intelligence can be used in the design of avatars to provide realistic responses to the user instead of basic animations with limited walking and arm movements. We hypothesise that providing smart avatars that can change

their facial expressions depending on the situation and/or be able to communicate with the participants can significantly enhance the persuasiveness of special cues in a VR environment and affect shoppers' behaviours.

In summary, the work has significant practical implications for the design and development of iVR environments, especially with respect to the use of virtual avatars and their effect on users' experienced presence dimensions. We believe our contribution paves the way for enhancing iVR environments, especially when used for creating unique shopping experiences. This can help retailers improve their consumer research practices and provide further opportunities for novel iVR e-commerce (v-commerce) platforms. Moreover, our work has significant practical implications for the design and development of iVR environments, especially with respect to the use of virtual avatars and their effect on users' experienced presence dimensions and user behavior.

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